

ELITE IGCSE ACADEMY

Private Past Paper Solutions

Nov 2025 P2H Worked Solutions

Nov 2025 | P2H

33 questions ■ question image followed by worked solution

PREPARED BY

Dr. Eslam Ahmed

Assistant Lecturer, Cairo University Faculty of Engineering

WhatsApp: +20 112 000 9622 | eliteigcse.com

TEACHER EDITION

Worked answers included

PRIVATE

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PAST PAPER QUESTION**Q#: 1****Marks: 2**TOPIC: **Coordinate Geometry**

Nov 2025 P2H - Nov 2025 | P2H

- 1** The point P has coordinates $(3, 4)$
The point Q has coordinates $(9, 16)$
 M is the midpoint of the line PQ
Find the coordinates of M

(..... ,)

(Total for Question 1 is 2 marks)

Dr. Eslam

PAST PAPER SOLUTION**Q#: 1****Marks: 2**TOPIC: **Coordinate Geometry**

Nov 2025 P2H - Nov 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Coordinate geometry. The tag is correct.**Solution**

$$P = (-3, 0), \quad Q = (9, 16)$$

The midpoint is

$$\begin{aligned} &\left(\frac{-3 + 9}{2}, \frac{0 + 16}{2} \right) \\ &= (3, 8) \end{aligned}$$

FINAL ANSWER $(3, 8).$

PAST PAPER QUESTION**Q#: 2****Marks: 3****TOPIC: Simultaneous Equations**

Nov 2025 P2H - Nov 2025 | P2H

2 Solve the simultaneous equations

$$5x + y = 11$$

$$3x - y = 9$$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

(Total for Question 2 is 3 marks)

PAST PAPER SOLUTION**Q#: 2****Marks: 3****TOPIC: Simultaneous Equations**

Nov 2025 P2H - Nov 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Simultaneous equations. The tag is correct.**Solution**

$$5x + y = 11$$

$$3x - y = 9$$

Add the equations:

$$8x = 20$$

$$x = 2.5$$

Substitute into $5x + y = 11$:

$$5(2.5) + y = 11$$

$$y = -1.5$$

FINAL ANSWER

$$x = 2.5, y = -1.5.$$

PAST PAPER QUESTION**Q#: 3****Marks: 4**TOPIC: **Fractions**

Nov 2025 P2H - Nov 2025 | P2H

3 A box contains only

9 red bricks
43 blue bricks
and some yellow bricks

$\frac{7}{20}$ of the bricks are yellow bricks.

Each brick weighs 35 grams.

Work out the total weight of the yellow bricks.

..... grams

(Total for Question 3 is 4 marks)

PAST PAPER SOLUTION**Q#: 3****Marks: 4**TOPIC: **Fractions**

Nov 2025 P2H - Nov 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Fractions. The tag is correct.**Method**

There are $9 + 43 = 52$ red and blue bricks.

Let the total number of bricks be T .

Since $\frac{7}{20}$ of the bricks are yellow, the remaining fraction is

$$1 - \frac{7}{20} = \frac{13}{20}$$

So

$$\frac{13}{20}T = 52$$

$$T = 52 \times \frac{20}{13} = 80$$

Yellow bricks:

$$\frac{7}{20} \times 80 = 28$$

Each brick weighs 35 grams, so total yellow-brick weight is

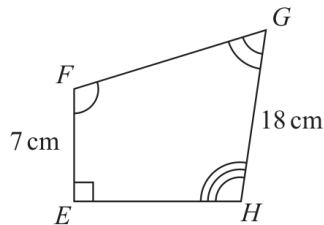
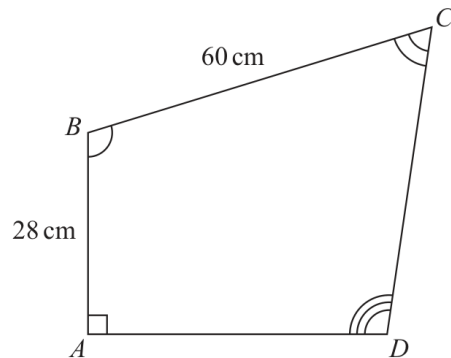
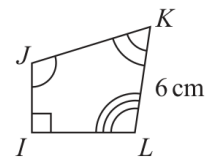
$$28 \times 35 = 980$$

FINAL ANSWER

980 grams

PAST PAPER QUESTION**Q#: 4** **Marks: 3****TOPIC:** Congruence, Similarity & Geometrical Proof

Nov 2025 P2H - Nov 2025 | P2H

4 Here are three similar quadrilaterals.Diagram **NOT**
accurately drawnWork out the length of JK

..... cm

(Total for Question 4 is 3 marks)

PAST PAPER SOLUTION**Q#: 4****Marks: 3****TOPIC:** Congruence, Similarity & Geometrical Proof

Nov 2025 P2H - Nov 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Congruence, Similarity & Geometrical Proof. The tag is correct.**Solution**

The scale factor from the largest quadrilateral to the middle quadrilateral is

$$\frac{7}{28} = \frac{1}{4}$$

So the side corresponding to 60 cm in the middle shape is

$$60 \times \frac{1}{4} = 15$$

The scale factor from the middle quadrilateral to the smallest quadrilateral is

$$\frac{6}{18} = \frac{1}{3}$$

$$JK = 15 \times \frac{1}{3} = 5$$

FINAL ANSWER

5 cm.

PAST PAPER QUESTION**Q#: 5****Marks: 6****TOPIC: Compound Interest & Depreciation**

Nov 2025 P2H - Nov 2025 | P2H

5 Otis sells ice creams.

On Friday, Otis sells 75 ice creams.

On Saturday, Otis sells 87 ice creams.

- (a) Work out the percentage increase in the number of ice creams Otis sells from Friday to Saturday.

..... %
(3)

Claudia buys an ice cream machine for 960 Swiss francs.

The value of the ice cream machine depreciates by 20% each year.

- (b) Work out the value of the ice cream machine at the end of 3 years.

..... Swiss francs
(3)

(Total for Question 5 is 6 marks)

PAST PAPER SOLUTION**Q#: 5****Marks: 6****TOPIC: Compound Interest & Depreciation**

Nov 2025 P2H - Nov 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed*

Topic check. Compound Interest and Depreciation. The tag is correct because part (b) is depreciation.

Method

For part (a), the increase is

$$87 - 75 = 12$$

Percentage increase:

$$\frac{12}{75} \times 100 = 16$$

For part (b), a depreciation of 20% gives multiplier

$$0.80$$

After 3 years,

$$960(0.80)^3 = 491.52$$

FINAL ANSWER

(a) 16%, (b) 491.52 Swiss francs

PAST PAPER QUESTION**Q#: 6****Marks: 5****TOPIC: Circles, Arcs & Sectors**

Nov 2025 P2H - Nov 2025 | P2H

- 6 A large circle with centre O contains 3 identical small circles.
Each circle touches two other circles.

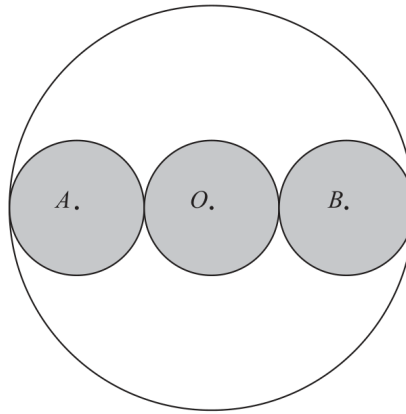


Diagram **NOT**
accurately drawn

A , O and B are the centres of the small circles.
 AOB is a straight line.

The circumference of the large circle is 160 cm

Work out the area of one small circle.
Give your answer correct to 3 significant figures.

..... cm²

(Total for Question 6 is 5 marks)

PAST PAPER SOLUTION**Q#: 6****Marks: 5**TOPIC: **Circles, Arcs & Sectors**

Nov 2025 P2H - Nov 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Corrected to circles, arcs and sectors.**Solution**

The circumference of the large circle is 160 cm, so

$$2\pi R = 160$$

$$R = \frac{80}{\pi}$$

The three equal small circles fit along the diameter of the large circle, so

$$2R = 6r$$

$$r = \frac{R}{3} = \frac{80}{3\pi}$$

Area of one small circle is

$$\begin{aligned}\pi r^2 &= \pi \left(\frac{80}{3\pi} \right)^2 \\ &= 226.35 \dots\end{aligned}$$

FINAL ANSWER226 cm² to 3 significant figures.

PAST PAPER QUESTION**Q#: 7****Marks: 3****TOPIC: Factorising**

Nov 2025 P2H - Nov 2025 | P2H

7 (i) Factorise $x^2 + 2x - 48$

.....
(2)

(ii) Hence, solve $x^2 + 2x - 48 = 0$

.....
(1)

(Total for Question 7 is 3 marks)

PAST PAPER SOLUTION**Q#: 7****Marks: 3**TOPIC: **Factorising**

Nov 2025 P2H - Nov 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Factorising. The tag is correct.**Solution**

$$x^2 + 2x - 48 = (x + 8)(x - 6)$$

So

$$(x + 8)(x - 6) = 0$$

FINAL ANSWER

$$x = -8 \text{ or } x = 6$$

PAST PAPER QUESTION**Q#: 8****Marks: 3****TOPIC: Powers, Roots & Standard Form**

Nov 2025 P2H - Nov 2025 | P2H

- 8 (a) Write 5.76×10^4 as an ordinary number.

.....
(1)

(b) Work out $\frac{3 \times 10^5 + 8 \times 10^3}{4 \times 10^{-2}}$

Give your answer in standard form.

.....
(2)**(Total for Question 8 is 3 marks)**

PAST PAPER SOLUTION**Q#: 8****Marks: 3****TOPIC: Powers, Roots & Standard Form**

Nov 2025 P2H - Nov 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Powers, roots and standard form. The tag is correct.**Method**

$$5.76 \times 10^4 = 57600$$

For part (b), simplify the numerator:

$$3 \times 10^5 + 8 \times 10^3 = 300000 + 8000 = 308000$$

Then

$$\frac{308000}{4 \times 10^{-2}} = \frac{308000}{0.04} = 7700000$$

In standard form:

$$7700000 = 7.7 \times 10^6$$

FINAL ANSWER(a) 57600, (b) 7.7×10^6

PAST PAPER QUESTION**Q#: 9****Marks: 3****TOPIC: Probability Toolkit**

Nov 2025 P2H - Nov 2025 | P2H

- 9 Mark has three bags, bag A, bag B and bag C

Bag A contains 120 counters.

The probability of taking at random a red counter from bag A is 0.45

Bag B contains 80 counters.

The probability of taking at random a red counter from bag B is 0.3

Bag C is empty.

Mark puts all the counters from bag A and bag B into bag C

Mark takes at random a counter from bag C

Work out the probability that he takes a red counter.

(Total for Question 9 is 3 marks)

PAST PAPER SOLUTION**Q#: 9****Marks: 3**TOPIC: **Probability Toolkit**

Nov 2025 P2H - Nov 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Probability Toolkit. The tag is correct.**Solution**

Bag A has

$$120 \times 0.45 = 54$$

red counters.

Bag B has

$$80 \times 0.30 = 24$$

red counters.

After combining the bags, the total number of red counters is

$$54 + 24 = 78$$

The total number of counters is

$$120 + 80 = 200$$

$$P(\text{red}) = \frac{78}{200} = 0.39$$

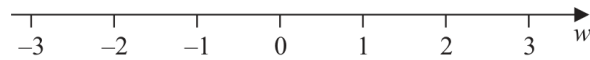
FINAL ANSWER

0.39.

PAST PAPER QUESTION**Q#: 10****Marks: 4****TOPIC: Graphing Inequalities**

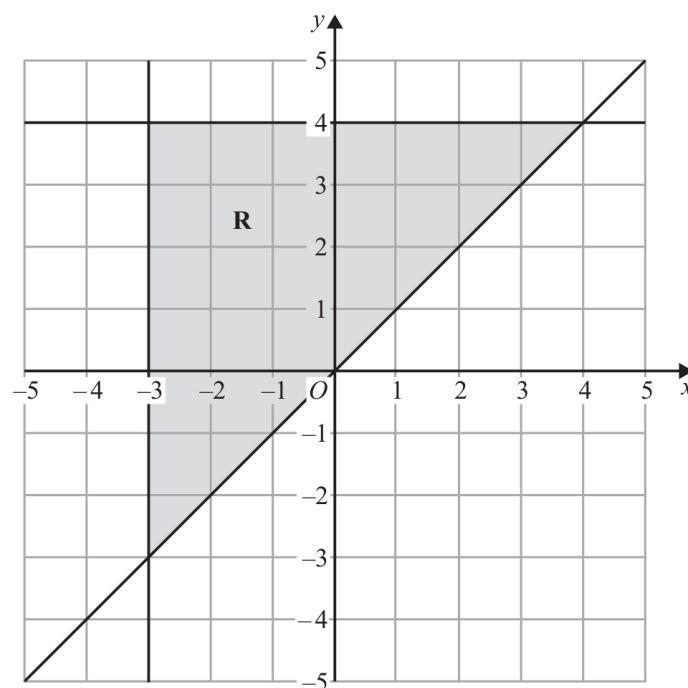
Nov 2025 P2H - Nov 2025 | P2H

- 10 (a) On the number line, represent the inequality $w < 1$



(1)

The region **R**, shown shaded in the diagram, is bounded by three straight lines.



- (b) Write down the three inequalities that define the region **R**

.....

(3)

(Total for Question 10 is 4 marks)

PAST PAPER SOLUTION**Q#: 10****Marks: 4****TOPIC: Graphing Inequalities**

Nov 2025 P2H - Nov 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct. This combines a number line and graph inequalities.**Solution**

For part (a), $w < 1$ is shown with an open circle at 1 and an arrow to the left.

For part (b), the region is right of $x = -3$, below $y = 4$, and above $y = x$:

$$x \geq -3$$

$$y \leq 4$$

$$y \geq x$$

FINAL ANSWER

(a) Open circle at 1, shade left. (b) $x \geq -3$, $y \leq 4$, $y \geq x$.

PAST PAPER QUESTION**Q#: 11****Marks: 4****TOPIC: Right-Angled Triangles - Pythagoras & Trigonometry**

Nov 2025 P2H - Nov 2025 | P2H

- 11** The diagram shows a right-angled triangle ABC

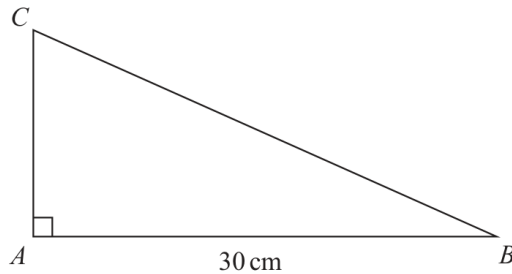


Diagram **NOT**
accurately drawn

Area of triangle $ABC = 240 \text{ cm}^2$

Work out the perimeter of triangle ABC

..... cm

(Total for Question 11 is 4 marks)

PAST PAPER SOLUTION**Q#: 11****Marks: 4****TOPIC: Right-Angled Triangles - Pythagoras & Trigonometry**

Nov 2025 P2H - Nov 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Right-Angled Triangles - Pythagoras & Trigonometry. The tag is correct.**Solution**Use the area to find AC :

$$\frac{1}{2} \times 30 \times AC = 240$$

$$15AC = 240$$

$$AC = 16$$

Then

$$BC^2 = 30^2 + 16^2 = 1156$$

$$BC = 34$$

Perimeter:

$$30 + 16 + 34 = 80$$

FINAL ANSWER

80 cm.

PAST PAPER QUESTION**Q#: 12****Marks: 5****TOPIC: Graphs of Functions**

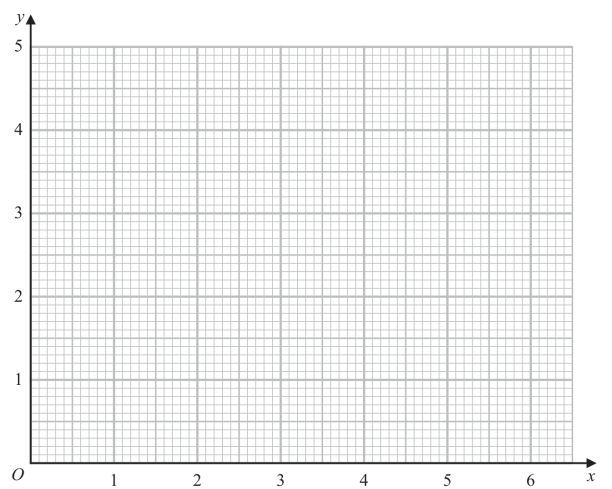
Nov 2025 P2H - Nov 2025 | P2H

12 (a) Complete the table of values for $y = \frac{1}{2}\left(x + \frac{4}{x}\right)$

x	0.5	1	2	3	4	5	6
y	4.25			2.17	2.5	2.9	3.33

(1)

(b) Draw the graph of $y = \frac{1}{2}\left(x + \frac{4}{x}\right)$ for $0.5 \leq x \leq 6$



(2)

(c) By drawing a suitable line on the grid, find estimates for the solutions of the equation $x + \frac{4}{x} = 6$

Give your answers correct to one decimal place.

(2)

(Total for Question 12 is 5 marks)

PAST PAPER SOLUTION**Q#: 12****Marks: 5****TOPIC: Graphs of Functions**

Nov 2025 P2H - Nov 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct. This is a reciprocal-curve graph and estimate question.**Solution**

$$y = \frac{1}{2} \left(x + \frac{4}{x} \right)$$

The completed table is:

x	0.5	1	2	3	4	5	6
y	4.25	2.5	2	2.17	2.5	2.9	3.33

For

$$x + \frac{4}{x} = 6$$

we need

$$\frac{1}{2} \left(x + \frac{4}{x} \right) = 3$$

So draw the horizontal line $y = 3$ on the graph.

The intersections give approximately

$$x = 0.8 \quad \text{and} \quad x = 5.2$$

FINAL ANSWER

Table missing values 2.5, 2. Estimates 0.8 and 5.2.

PAST PAPER QUESTION**Q#: 13****Marks: 4**TOPIC: **Algebraic Roots & Indices**

Nov 2025 P2H - Nov 2025 | P2H

13 (a) Simplify fully $\left(\frac{125x^6}{y^{15}}\right)^{\frac{1}{3}}$

.....
(2)

Given that $4^n = \frac{4^m}{64^p}$

(b) Express n in terms of m and p

$n =$
(2)

(Total for Question 13 is 4 marks)

PAST PAPER SOLUTION**Q#: 13****Marks: 4**TOPIC: **Algebraic Roots & Indices**

Nov 2025 P2H - Nov 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Algebraic roots and indices. The tag is correct.**Solution**

(a)

$$\left(\frac{125x^6}{y^{15}}\right)^{1/3} = \frac{5x^2}{y^5}$$

(b)

$$64^p = (4^3)^p = 4^{3p}$$

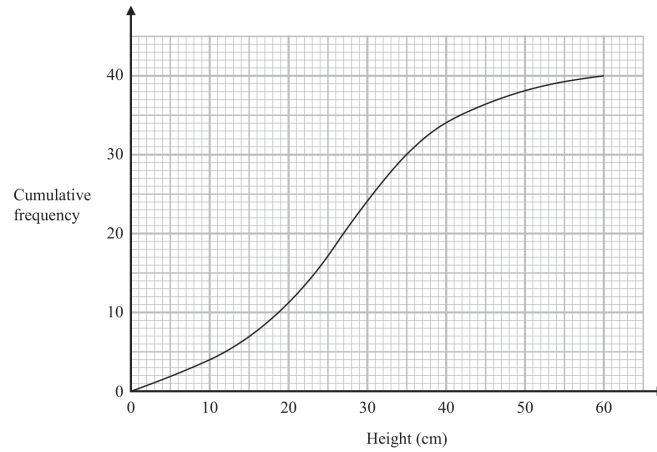
$$4^n = \frac{4^m}{64^p} = \frac{4^m}{4^{3p}} = 4^{m-3p}$$

FINAL ANSWER(a) $\frac{5x^2}{y^5}$, (b) $n = m - 3p$

PAST PAPER QUESTION**Q#: 14****Marks: 5****TOPIC: Cumulative Frequency Diagrams**

Nov 2025 P2H - Nov 2025 | P2H

- 14 The cumulative frequency graph shows information about the heights of 40 plants that Greta has grown.



- (a) Use the graph to find an estimate for the median height.

..... cm
(1)

- (b) Use the graph to find an estimate for the interquartile range of the heights.

..... cm
(2)

Plants with a height greater than 40 cm are premium plants.
Greta sells all the premium plants for 30 euros each.

- (c) Work out the total amount of money Greta receives for the premium plants.

..... euros
(2)

(Total for Question 14 is 5 marks)

PAST PAPER SOLUTION**Q#: 14****Marks: 5****TOPIC: Cumulative Frequency Diagrams**

Nov 2025 P2H - Nov 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Cumulative Frequency Diagrams. The tag is correct.**Solution**

There are 40 plants.

From the graph,

$$\text{median} \approx 26 \text{ cm}$$

$$Q_1 \approx 19 \text{ cm}, \quad Q_3 \approx 35 \text{ cm}$$

$$\text{IQR} \approx 35 - 19 = 16 \text{ cm}$$

For premium plants, use height > 40 cm.

$$F(40) \approx 34$$

$$40 - 34 = 6$$

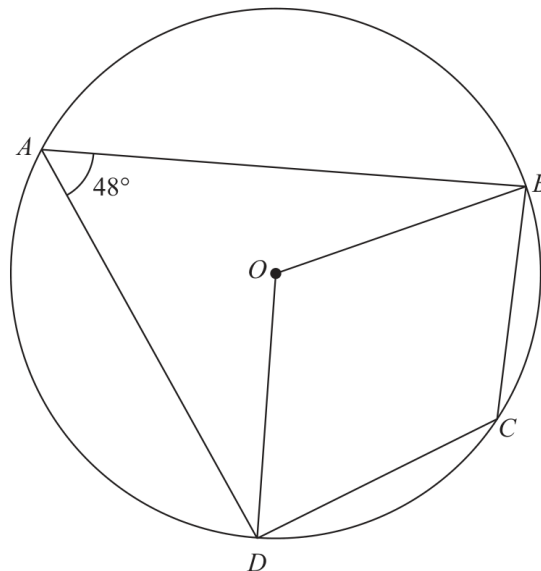
$$6 \times 30 = 180$$

FINAL ANSWER

median about 26 cm, IQR about 16 cm, and 180 euros.

PAST PAPER QUESTION**Q#: 15****Marks: 4**TOPIC: **Circle Theorems**

Nov 2025 P2H - Nov 2025 | P2H

15 A, B, C and D are points on a circle with centre O Diagram **NOT**
accurately drawnAngle $DAB = 48^\circ$ (a) (i) Work out the size of the obtuse angle DOB
(1)

(ii) Give a reason for your answer.

.....
(1)(b) (i) Work out the size of angle BCD
(1)

(ii) Give a reason for your answer.

.....
(1)**(Total for Question 15 is 4 marks)**

PAST PAPER SOLUTION**Q#: 15****Marks: 4****TOPIC: Circle Theorems**

Nov 2025 P2H - Nov 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Circle Theorems. The tag is correct.**Solution**

(a) The angle at the centre is twice the angle at the circumference standing on chord DB :

$$\angle DOB = 2 \times 48^\circ = 96^\circ$$

(b) Opposite angles in cyclic quadrilateral $ABCD$ add to 180° :

$$\angle BCD = 180^\circ - 48^\circ = 132^\circ$$

FINAL ANSWER $\angle DOB = 96^\circ$, and $\angle BCD = 132^\circ$.

PAST PAPER QUESTION**Q#: 16****Marks: 2**TOPIC: **Algebraic Proof**

Nov 2025 P2H - Nov 2025 | P2H

16 Use algebra to show that $0.\dot{3}\dot{7}\dot{8} = \frac{25}{66}$

(Total for Question 16 is 2 marks)

PAST PAPER SOLUTION**Q#: 16****Marks: 2**TOPIC: **Algebraic Proof**

Nov 2025 P2H - Nov 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Algebraic proof. The tag is correct.**Solution**

Let

$$x = 0.3787878 \dots$$

Then

$$10x = 3.787878 \dots$$

and

$$1000x = 378.787878 \dots$$

Subtract:

$$990x = 375$$

$$x = \frac{375}{990} = \frac{25}{66}$$

FINAL ANSWER

$$0.3787878 \dots = \frac{25}{66}.$$

PAST PAPER QUESTION**Q#: 17****Marks: 3****TOPIC: Direct & Inverse Proportion**

Nov 2025 P2H - Nov 2025 | P2H

17 T is inversely proportional to $\sqrt{m}$

$T = 15$ when $m = 36$

Find a formula for T in terms of m

.....
(Total for Question 17 is 3 marks)

PAST PAPER SOLUTION**Q#: 17****Marks: 3****TOPIC: Direct & Inverse Proportion**

Nov 2025 P2H - Nov 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Direct and Inverse Proportion. The tag is correct.**Method** T is inversely proportional to m , so

$$T = \frac{k}{m}$$

Use $T = 15$ when $m = 36$:

$$15 = \frac{k}{36}$$

$$k = 540$$

Therefore

$$T = \frac{540}{m}$$

FINAL ANSWER

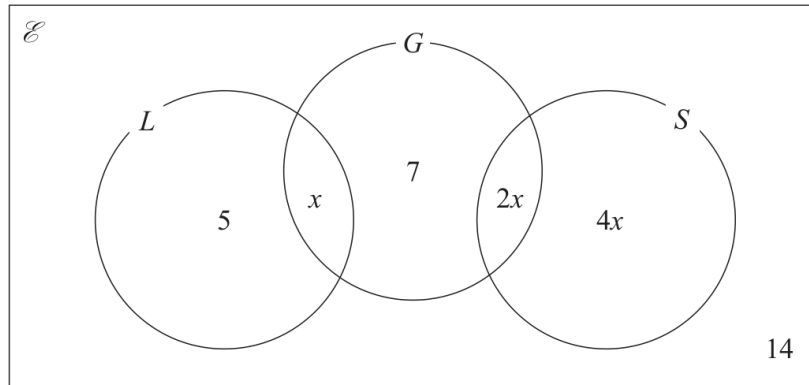
$$T = \frac{540}{m}.$$

PAST PAPER QUESTION**Q#: 18****Marks: 3****TOPIC: Probability Diagrams - Venn & Tree Diagrams**

Nov 2025 P2H - Nov 2025 | P2H

- 18** Bianca asked 40 students which of the languages Latin (L), Greek (G) and Sanskrit (S) they study.

The Venn diagram gives some information about her results.
It shows the number of students in each subset.



One of these students is selected at random.
Work out the probability that this student studies Latin or Sanskrit.

(Total for Question 18 is 3 marks)

PAST PAPER SOLUTION**Q#: 18****Marks: 3****TOPIC: Probability Diagrams - Venn & Tree Diagrams**

Nov 2025 P2H - Nov 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Probability Diagrams - Venn & Tree Diagrams. The tag is correct.**Solution**

The total is 40.

The regions shown add to

$$5 + x + 7 + 2x + 4x + 14$$

So

$$26 + 7x = 40$$

$$7x = 14$$

$$x = 2$$

Latin or Sanskrit:

$$5 + x + 2x + 4x = 5 + 2 + 4 + 8 = 19$$

$$P(L \cup S) = \frac{19}{40}$$

FINAL ANSWER

$$\frac{19}{40}$$

PAST PAPER QUESTION**Q#: 19****Marks: 3**TOPIC: **Functions**

Nov 2025 P2H - Nov 2025 | P2H

19 The functions f and g are such that

$$f(x) = 3x - 2$$

$$g(x) = \frac{x}{2x - 1}$$

(a) Find $g(3)$

(1)(b) Find $gf(x)$

Give your answer in its simplest form.

$$gf(x) = \dots\dots\dots$$

(2)

(Total for Question 19 is 3 marks)

PAST PAPER SOLUTION**Q#: 19****Marks: 3****TOPIC: Functions**

Nov 2025 P2H - Nov 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Functions. The tag is correct.**Solution**

$$f(x) = 3x - 2, \quad g(x) = \frac{x}{x - 1}$$

(a)

$$g(3) = \frac{3}{3 - 1} = \frac{3}{2}$$

(b)

$$gf(x) = g(f(x))$$

$$gf(x) = \frac{3x - 2}{(3x - 2) - 1}$$

$$gf(x) = \frac{3x - 2}{3x - 3}$$

FINAL ANSWER

$$g(3) = \frac{3}{2}, \text{ and } gf(x) = \frac{3x - 2}{3x - 3}.$$

PAST PAPER QUESTION**Q#: 20****Marks: 3****TOPIC: Forming & Solving Equations**

Nov 2025 P2H - Nov 2025 | P2H

20 Here is a quadratic equation.

$$ax^2 + 4x + c = 0$$

The solutions of this equation are given by $x = \frac{-4 \pm 2\sqrt{39}}{10}$

Find the value of a and the value of c
Show your working clearly.

 $a = \dots\dots\dots$ $c = \dots\dots\dots$

(Total for Question 20 is 3 marks)

PAST PAPER SOLUTION**Q#: 20****Marks: 3****TOPIC: Forming & Solving Equations**

Nov 2025 P2H - Nov 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct.**Method**

For

$$ax^2 + 4x + c = 0$$

the quadratic formula gives

$$x = \frac{-4 \pm \sqrt{4^2 - 4ac}}{2a}$$

The given solutions are

$$x = \frac{-4 \pm 2\sqrt{39}}{10}$$

Compare denominators:

$$2a = 10$$

$$a = 5$$

Compare the square root parts:

$$\sqrt{16 - 4ac} = 2\sqrt{39}$$

Square both sides:

$$16 - 4ac = 156$$

Use $a = 5$:

$$16 - 20c = 156$$

$$-20c = 140$$

$$c = -7$$

FINAL ANSWER

$$a = 5, c = -7$$

PAST PAPER QUESTION**Q#: 20****Marks: 3****TOPIC: Forming & Solving Equations**

Nov 2025 P2H - Nov 2025 | P2H

20 Here is a quadratic equation.

$$ax^2 + 4x + c = 0$$

The solutions of this equation are given by $x = \frac{-4 \pm 2\sqrt{39}}{10}$

Find the value of a and the value of c
Show your working clearly.

 $a = \dots\dots\dots$ $c = \dots\dots\dots$

(Total for Question 20 is 3 marks)

PAST PAPER SOLUTION**Q#: 20****Marks: 3****TOPIC: Forming & Solving Equations**

Nov 2025 P2H - Nov 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct.**Method**

For

$$ax^2 + 4x + c = 0$$

the quadratic formula gives

$$x = \frac{-4 \pm \sqrt{4^2 - 4ac}}{2a}$$

The given solutions are

$$x = \frac{-4 \pm 2\sqrt{39}}{10}$$

Compare denominators:

$$2a = 10$$

$$a = 5$$

Compare the square root parts:

$$\sqrt{16 - 4ac} = 2\sqrt{39}$$

Square both sides:

$$16 - 4ac = 156$$

Use $a = 5$:

$$16 - 20c = 156$$

$$-20c = 140$$

$$c = -7$$

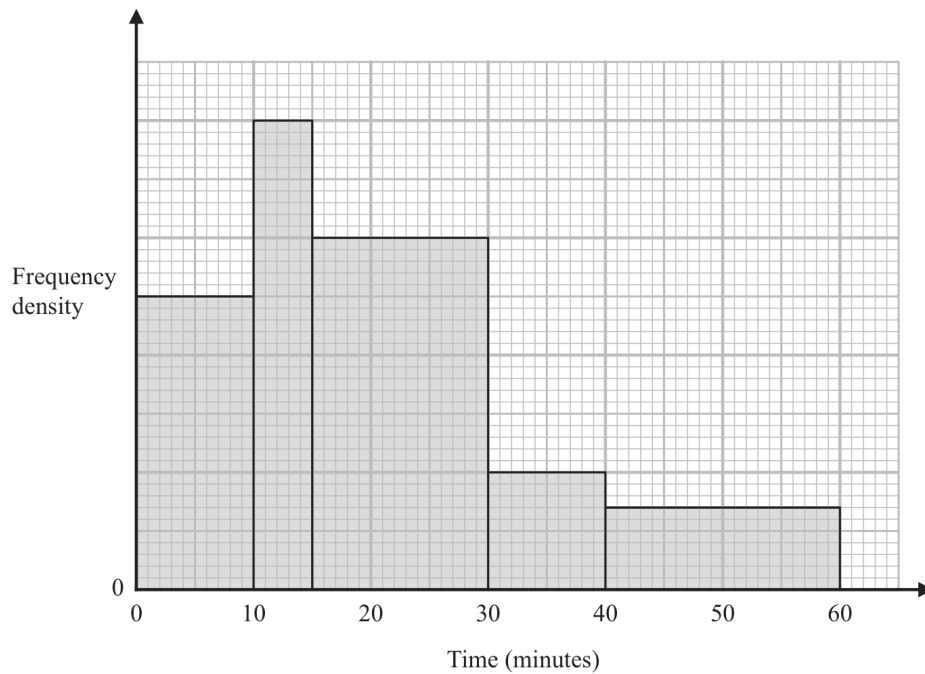
FINAL ANSWER

$$a = 5, c = -7$$

PAST PAPER QUESTION**Q#: 21****Marks: 3****TOPIC: Histograms**

Nov 2025 P2H - Nov 2025 | P2H

- 21** The histogram shows information about the times, in minutes, that some trains arrived late at a station one day.



20 of these trains arrived between 10 minutes late and 15 minutes late.

No trains arrived more than 60 minutes late.

Work out an estimate for the number of these trains that arrived at least 25 minutes late.

(Total for Question 21 is 3 marks)

PAST PAPER SOLUTION**Q#: 21****Marks: 3**TOPIC: **Histograms**

Nov 2025 P2H - Nov 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct.**Method**

For a histogram,

$$\text{frequency} = \text{class width} \times \text{frequency density}$$

For $10 < t \leq 15$, the frequency is 20 and the class width is 5, so

$$\text{frequency density} = \frac{20}{5} = 4$$

Using this scale from the histogram:

$$15 < t \leq 30 \quad \text{has frequency density } 3$$

$$30 < t \leq 40 \quad \text{has frequency density } 1$$

$$40 < t \leq 60 \quad \text{has frequency density } 0.7$$

For trains at least 25 minutes late:

$$(5)(3) + (10)(1) + (20)(0.7)$$

$$= 15 + 10 + 14$$

$$= 39$$

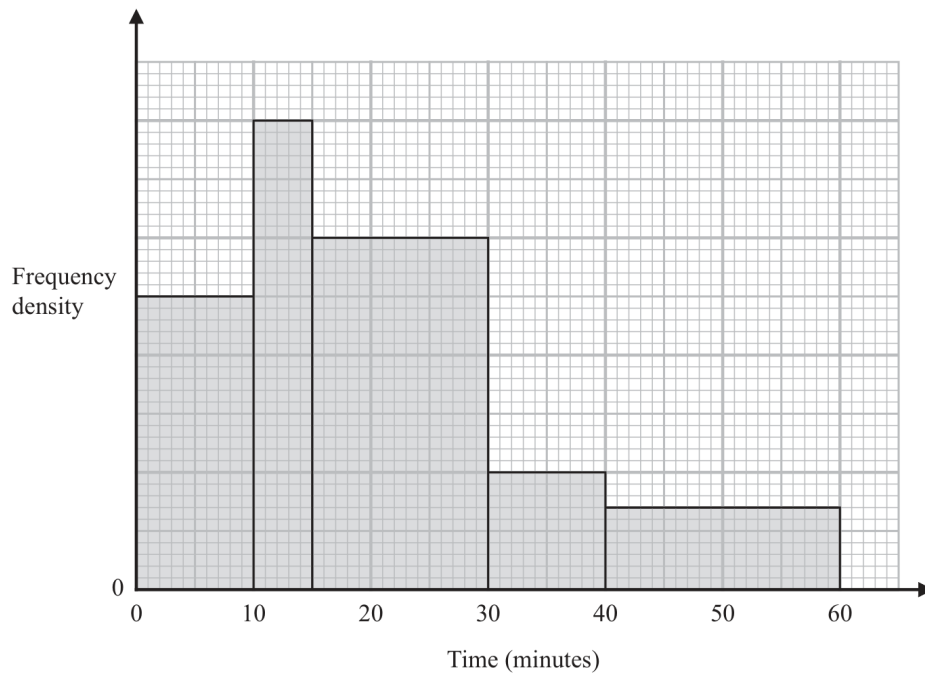
FINAL ANSWER

39 trains

PAST PAPER QUESTION**Q#: 21****Marks: 3****TOPIC: Histograms**

Nov 2025 P2H - Nov 2025 | P2H

- 21** The histogram shows information about the times, in minutes, that some trains arrived late at a station one day.



20 of these trains arrived between 10 minutes late and 15 minutes late.

No trains arrived more than 60 minutes late.

Work out an estimate for the number of these trains that arrived at least 25 minutes late.

(Total for Question 21 is 3 marks)

PAST PAPER SOLUTION**Q#: 21****Marks: 3**TOPIC: **Histograms**

Nov 2025 P2H - Nov 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct.**Method**

For a histogram,

$$\text{frequency} = \text{class width} \times \text{frequency density}$$

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For trains at least 25 minutes late:

$$(5)(3) + (10)(1) + (20)(0.7)$$

$$= 15 + 10 + 14$$

$$= 39$$

FINAL ANSWER

39 trains

PAST PAPER QUESTION**Q#: 22****Marks: 6****TOPIC: Differentiation**

Nov 2025 P2H - Nov 2025 | P2H

22 Curve **C** has equation $y = x^3 - 16x + 7$

At two points on **C**, the gradient is 11

The tangents to **C** at these two points have equations of the form $y = ax + b$

Work out the two possible values of b

Show clear algebraic working.

.....
(Total for Question 22 is 6 marks)

PAST PAPER SOLUTION**Q#: 22****Marks: 6****TOPIC: Differentiation**

Nov 2025 P2H - Nov 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Moved to Differentiation.**Method**

$$y = x^3 - 16x + 7$$

Differentiate:

$$\frac{dy}{dx} = 3x^2 - 16$$

The gradient of the tangent is 11, so

$$3x^2 - 16 = 11$$

$$3x^2 = 27$$

$$x^2 = 9$$

$$x = 3 \quad \text{or} \quad x = -3$$

When $x = 3$:

$$y = 3^3 - 16(3) + 7 = 27 - 48 + 7 = -14$$

The tangent has equation $y = 11x + b$, so

$$-14 = 11(3) + b$$

$$b = -47$$

When $x = -3$:

$$y = (-3)^3 - 16(-3) + 7 = -27 + 48 + 7 = 28$$

$$28 = 11(-3) + b$$

$$b = 61$$

FINAL ANSWER

$$b = -47 \text{ or } b = 61$$

PAST PAPER QUESTION**Q#: 22****Marks: 6****TOPIC: Differentiation**

Nov 2025 P2H - Nov 2025 | P2H

22 Curve **C** has equation $y = x^3 - 16x + 7$

At two points on **C**, the gradient is 11

The tangents to **C** at these two points have equations of the form $y = ax + b$

Work out the two possible values of b

Show clear algebraic working.

.....
(Total for Question 22 is 6 marks)

PAST PAPER SOLUTION**Q#: 22****Marks: 6****TOPIC: Differentiation**

Nov 2025 P2H - Nov 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Moved to Differentiation.**Method**

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$$y = (-3)^3 - 16(-3) + 7 = -27 + 48 + 7 = 28$$

$$28 = 11(-3) + b$$

$$b = 61$$

FINAL ANSWER

$$b = -47 \text{ or } b = 61$$

PAST PAPER QUESTION**Q#: 23****Marks: 3****TOPIC: Area & Volume of Similar Shapes**

Nov 2025 P2H - Nov 2025 | P2H

23 Shape **P** is similar to shape **Q**The table shows some information about shape **P** and shape **Q**

	Surface area (cm ²)	Volume (cm ³)
Shape P	200	672
Shape Q	450	

Work out the volume of shape **Q**..... cm³**(Total for Question 23 is 3 marks)**

PAST PAPER SOLUTION**Q#: 23****Marks: 3****TOPIC: Area & Volume of Similar Shapes**

Nov 2025 P2H - Nov 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct.**Method**

Surface area ratio Q:P:

$$450 : 200 = 9 : 4$$

So the linear ratio Q:P is

$$3 : 2$$

Therefore the volume ratio Q:P is

$$3^3 : 2^3 = 27 : 8$$

Shape P has volume 672 cm^3 .

Volume of Q:

$$672 \times \frac{27}{8} = 2268$$

FINAL ANSWER**2268 cm³**

PAST PAPER QUESTION**Q#: 23****Marks: 3****TOPIC: Area & Volume of Similar Shapes**

Nov 2025 P2H - Nov 2025 | P2H

23 Shape **P** is similar to shape **Q**The table shows some information about shape **P** and shape **Q**

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Shape P	200	672
Shape Q	450	

Work out the volume of shape **Q**..... cm³**(Total for Question 23 is 3 marks)**

PAST PAPER SOLUTION**Q#: 23****Marks: 3****TOPIC: Area & Volume of Similar Shapes**

Nov 2025 P2H - Nov 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct.**Method**

Surface area ratio Q:P:

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Therefore the volume ratio Q:P is

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Shape P has volume 672 cm^3 .

Volume of Q:

$$672 \times \frac{27}{8} = 2268$$

FINAL ANSWER**2268 cm³**

PAST PAPER QUESTION**Q#: 24****Marks: 5****TOPIC: Rounding, Estimation & Bounds**

Nov 2025 P2H - Nov 2025 | P2H

24 $D = \frac{n}{p - q}$

 $n = 10.3$ correct to 1 decimal place $p = 7.24$ correct to 2 decimal places $q = 4.39$ correct to 2 decimal places

By considering bounds, work out the value of D to a suitable degree of accuracy.
Show your working clearly.

(Total for Question 24 is 5 marks)

PAST PAPER SOLUTION**Q#: 24****Marks: 5****TOPIC: Rounding, Estimation & Bounds**

Nov 2025 P2H - Nov 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Rounding, Estimation and Bounds. The tag is correct.**Method**

$$D = \frac{n}{p - q}$$

Bounds:

$$10.25 \leq n < 10.35$$

$$7.235 \leq p < 7.245$$

$$4.385 \leq q < 4.395$$

Lower bound:

$$D_{\text{lower}} = \frac{10.25}{7.245 - 4.385} = 3.5839 \dots$$

Upper bound:

$$D_{\text{upper}} = \frac{10.35}{7.235 - 4.395} = 3.6443 \dots$$

So

$$3.5839 \dots < D < 3.6443 \dots$$

Both bounds round to 3.6 to 1 decimal place.

FINAL ANSWER $D = 3.6$ to a suitable degree of accuracy

PAST PAPER QUESTION**Q#: 24****Marks: 5****TOPIC: Rounding, Estimation & Bounds**

Nov 2025 P2H - Nov 2025 | P2H

24 $D = \frac{n}{p - q}$

 $n = 10.3$ correct to 1 decimal place $p = 7.24$ correct to 2 decimal places $q = 4.39$ correct to 2 decimal places

By considering bounds, work out the value of D to a suitable degree of accuracy.
Show your working clearly.

(Total for Question 24 is 5 marks)

PAST PAPER SOLUTION**Q#: 24****Marks: 5****TOPIC: Rounding, Estimation & Bounds**

Nov 2025 P2H - Nov 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Rounding, Estimation & Bounds is correct.**Method**

$$D = \frac{n}{p - q}$$

Bounds:

$$10.25 < n < 10.35$$

$$7.235 \leq p < 7.245$$

$$4.385 \leq q < 4.395$$

Lower bound:

$$\frac{10.25}{7.245 - 4.385} = 3.5839...$$

Upper bound:

$$\frac{10.35}{7.235 - 4.395} = 3.6443...$$

So $3.5839... \leq D < 3.6443...$ Both bounds round to 3.6 to 1 decimal place.**FINAL ANSWER** $D = 3.6$ to a suitable degree of accuracy.

PAST PAPER QUESTION**Q#: 25****Marks: 5**TOPIC: **Sequences**

Nov 2025 P2H - Nov 2025 | P2H

- 25** The first term of an arithmetic series is 10
The 20th term of the series is 86
The sum of the first N terms of the series is 5194
Work out the value of N
Show your working clearly.

 $N = \dots\dots\dots$

(Total for Question 25 is 5 marks)

PAST PAPER SOLUTION**Q#: 25****Marks: 5**TOPIC: **Sequences**

Nov 2025 P2H - Nov 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct.**Method**

$$a = 10$$

$$u_{20} = 86$$

$$10 + 19d = 86$$

$$d = 4$$

Use

$$S_N = \frac{N}{2}(2a + (N - 1)d)$$

$$5194 = \frac{N}{2}(20 + 4(N - 1))$$

$$5194 = \frac{N}{2}(4N + 16)$$

$$5194 = 2N(N + 4)$$

$$N^2 + 4N - 2597 = 0$$

$$N = \frac{-4 + 102}{2} = 49$$

FINAL ANSWER

$$N = 49$$

PAST PAPER QUESTION**Q#: 25****Marks: 5**TOPIC: **Sequences**

Nov 2025 P2H - Nov 2025 | P2H

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The 20th term of the series is 86

The sum of the first N terms of the series is 5194

Work out the value of N
Show your working clearly.

 $N = \dots\dots\dots$

(Total for Question 25 is 5 marks)

PAST PAPER SOLUTION**Q#: 25****Marks: 5**TOPIC: **Sequences**

Nov 2025 P2H - Nov 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct.**Method**

$$a = 10$$

$$u_{20} = 86$$

$$10 + 19d = 86$$

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Use

$$S_N = \frac{N}{2}(2a + (N - 1)d)$$

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$$5194 = \frac{N}{2}(4N + 16)$$

$$5194 = 2N(N + 4)$$

$$N^2 + 4N - 2597 = 0$$

$$N = \frac{-4 + 102}{2} = 49$$

FINAL ANSWER

$$N = 49$$

PAST PAPER QUESTION**Q#: 26****Marks: 6****TOPIC: Standard & Compound Units**

Nov 2025 P2H - Nov 2025 | P2H

26 A solid cone is joined to a solid hemisphere to make the solid shown below.

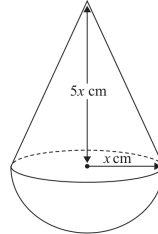


Diagram NOT
accurately drawn

The cone is made from copper.
The density of copper is 9 g/cm^3

The hemisphere is made from a different metal.

The total mass of the solid is 4752π grams

The total volume of the solid is $504\pi \text{ cm}^3$

Work out the density of the hemisphere.
Show your working clearly.

..... g/cm^3

(Total for Question 26 is 6 marks)

PAST PAPER SOLUTION**Q#: 26****Marks: 6****TOPIC: Standard & Compound Units**

Nov 2025 P2H - Nov 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct.**Method**

Volume of cone:

$$\frac{1}{3}\pi x^2(5x) = \frac{5}{3}\pi x^3$$

Volume of hemisphere:

$$\frac{2}{3}\pi x^3$$

Total volume:

$$\frac{5}{3}\pi x^3 + \frac{2}{3}\pi x^3 = 504\pi$$

$$\frac{7}{3}\pi x^3 = 504\pi$$

$$x^3 = 216$$

$$x = 6$$

Cone volume:

$$\frac{5}{3}\pi(6)^3 = 360\pi$$

Mass of copper cone:

$$360\pi \times 9 = 3240\pi$$

Mass of hemisphere:

$$4752\pi - 3240\pi = 1512\pi$$

Hemisphere volume:

$$\frac{2}{3}\pi(6)^3 = 144\pi$$

Density of the hemisphere:

$$\frac{1512\pi}{144\pi} = 10.5$$

FINAL ANSWER

$$10.5 \text{ g/cm}^3$$

PAST PAPER QUESTION**Q#: 26****Marks: 6****TOPIC: Standard & Compound Units**

Nov 2025 P2H - Nov 2025 | P2H

26 A solid cone is joined to a solid hemisphere to make the solid shown below.

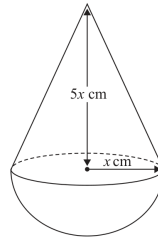


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The hemisphere is made from a different metal.

The total mass of the solid is 4752π grams

The total volume of the solid is $504\pi \text{ cm}^3$

Work out the density of the hemisphere.
Show your working clearly.

..... g/cm^3

(Total for Question 26 is 6 marks)

PAST PAPER SOLUTION**Q#: 26****Marks: 6****TOPIC: Standard & Compound Units**

Nov 2025 P2H - Nov 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct.**Method**

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Volume of hemisphere:

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$$\frac{5}{3}\pi(6)^3 = 360\pi$$

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Mass of hemisphere:

$$4752\pi - 3240\pi = 1512\pi$$

Hemisphere volume:

$$\frac{2}{3}\pi(6)^3 = 144\pi$$

Density of the hemisphere:

$$\frac{1512\pi}{144\pi} = 10.5$$

FINAL ANSWER

$$10.5 \text{ g/cm}^3$$